

StraTracker: A dynamic counting method for growing strawberries based on multi-target tracking

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ABSTRACT

Accurately counting fruit in orchards is a critical step for effective digital farming management. However, the variability in fruit size, overlapping shadows, and light interference present significant challenges to applying computer vision during the strawberry growth phase. To address these challenges, we propose StraTracker, a multi-object tracking (MOT) algorithm specifically designed to identify and count strawberries at various growth stages. StraTracker transforms the counting task into a frame-by-frame tracking problem, integrating both motion and appearance features. The algorithm is composed of three key components: a strawberry detector based on YOLOv8n, a feature association module, and a dual-area counting (DC) module. First, the strawberry detector accurately recognizes five growth stages, achieving an average accuracy of 91.93 % at 38.3 FPS. Next, the feature association module, incorporating the Feature Slicing Attention (FSA) and Adaptive Kalman Filtering (AKF) modules, mitigates issues such as light interference, impractical tracking frames, and ID switching (IDs). As a result, StraTracker achieves a Multi-Object Tracking Accuracy (MOTA) of 83.28 % and a Higher-Order Tracking Accuracy (HOTA) of 77.26 %, with only 259 IDs, outperforming existing baseline models. Finally, the DC module categorizes fruit counts based on the unique IDs assigned during tracking. The algorithm's coefficient of determination ($R^2 = 0.91$) and GEH of 2.33 indicate a strong correlation between predicted and actual counts. In conclusion, StraTracker offers a promising solution for farmers to optimize planting strategies and develop more precise harvesting plans.

1. Introduction

Strawberries are a crop of great economic and nutritional importance, with global cultivation expanding from 310,000 ha in 1999 to more than 900,000 ha by 2023 (Häkkinen et al., 2000; Gudowska et al., 2024). However, the absence of timely monitoring frequently results in challenges such as incomplete flower thinning, untimely picking, and improper fertilization, significantly diminishing fruit production (Samtani et al., 2019). Therefore, real-time solutions for monitoring fruit growth stages and counting information are urgently needed. Some existing monitoring solutions rely on manual observation or specialized instruments (Häni et al., 2020; Culman et al., 2023), but these are expensive and require damage to the fruit (Scott et al., 2021). In contrast, deep learning and computer vision-based methods provide a new solution for digital farming management with the benefits of low cost, high efficiency, and real-time performance (Attri et al., 2023).

Deep learning addresses the challenge of target recognition and detection by modeling human visual perception (Sanaeifar et al., 2023).

The orchard has unique challenges compared to general detection scenarios, including complex backgrounds, diverse fruit growth patterns, and light interference (Lu et al., 2023). These factors increase the complexity of fruit counting, which leads to most of the studies improving the accuracy of recognition and counting by improving the models of target detection (Vasconez et al., 2020). For example, Song et al. (2014) employed a color classifier to count chili fruits in images, achieving a correlation of 74 % with manual counting. Mirhaji et al. (2021) introduced the utilization of the YOLO model for detecting and counting ripe oranges in an orchard, achieving a detection accuracy mAP of 90.8 %. Parvathi et al. (2021) utilized Fast-RCNN and ResNet-50 algorithmic models to assess the ripeness of coconuts, classifying them into two categories: ripe or unripe. Bhattarai et al. (2022) employed weakly supervised methods to detect apple flowers and fruits, achieving root mean square errors of 2.9 and 4.3, respectively. However, traditional target detection encounters significant limitations in monitoring strawberry growth: it can only effectively analyze static single-frame images, rendering it inadequate for the comprehensive task of

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counting and overseeing the entire orchard. To overcome this challenge, we employ Multi-Object Tracking (MOT) to facilitate holistic orchard monitoring through cross-frame correlation.

MOT algorithms leverage information from preceding and succeeding frames to predict the target's state, thus facilitating cross-frame target association (Y. Zhang et al., 2024). The most canonical and widely-adopted MOT algorithms include the SORT (Bewley et al., 2016), DeepSORT (Pujara et al., 2022), and BoT-SORT (Aharon et al., 2022) models. Both SORT and DeepSORT algorithms utilize KF to forecast target trajectories and employ an IOU model for data similarity matching. DeepSORT further incorporates a ReID module to re-extract target appearance features. BoT-SORT, recognized as a state-of-the-art (SOTA) MOT algorithm, augments the feature association module by integrating motion and appearance information, implementing a camera motion compensation module, and optimizing the model structure with a low-score quadratic correlation function.

In the field of agriculture, researchers track and count fruits using UAVs (Zheng et al., 2023), Robots (Mirbod et al., 2023), or Monitoring platforms (W. L. Zhang et al., 2022). For example, He et al. (2022) monitored and counted camellia flowers using YOLOv3, Kalman filtering (KF), and multiple feature matching through video captured by a cell phone, achieving a MOTA of 66.1 %. Gao et al. (2022) used KF and Euclidean distance to count ripe apples in robotic applications, achieving 69.14 % MOTA with their algorithm. Lin et al. (2022) integrated the improved YOLOv5s and Deepsort algorithms to count peanuts on UAVs. This approach assigns ID to all targets using the MOT algorithm, facilitating counting by tracking these ID. However, algorithmic errors can cause ID switching (IDs), which exaggerates the number of fruits by resulting in mismatched counts. Wu et al. (2023) developed an automated fruit counting system for apples on mobile platforms by combining abnormal fruit detection, trunk tracking, and ID assignment, achieving a MOTA of 89.9 %. Bolouri et al. (2024) combined Mask-RCNN with SORT for the split counting of cotton in four states at the edge device. Although existing studies have effectively integrated target detection and tracking algorithms, they are limited to fruit targets ripening during the same time period, such as camellia, peanuts and apples. These studies lack the capability to comprehensively monitor the entire growth cycle of strawberries. Specifically, strawberries continue to produce new flowers and fruits throughout their growth cycle, resulting in the simultaneous appearance of fruits at different growth

stages (Yang et al., 2010; J. Zhang et al., 2023). Hence, the MOT method applied to strawberry fruits during the growing stages requires further improvement and modification. Additionally, the SOTA BoT-SORT algorithm still faces challenges related to tracking frame irrelevance, light interference, and IDs.

To address the above challenges, we propose a dynamic counting model for strawberries named StraTracker based on the BoT-SORT algorithm. The structural model structure is illustrated in Fig. 1. To the best of our knowledge, this is the inaugural study of the MOT algorithm for monitoring and counting fruits during the strawberry growth cycle. The StraTracker model monitors and visualizes the distribution of fruit quantities in the orchard, which facilitates timely management and harvesting by growers. The main contributions of this paper can be summarized as follows:

Collected and curated a StraMOT dataset tailored for the MOT algorithm, undergoing data calibration and preprocessing.
Designed the Feature Slice Attention (FSA) module to refine small target features and enhance the real-time performance of the tracking model.

Introduced the Adaptive Kalman Filter (AKF) module to optimise the ability of model adaptive tuning under real-time scene changes.

Implemented a Dual-area Counting method (DC) at the output to quantify the number of fruits based on their unique identifiers (ID).

This work is structured as follows: Section 2 describes the methodology for data collection and preprocessing. Section 3 outlines the methodology, highlighting improvements to the detector, feature association, and counting modules. Section 4 details the experiments conducted, providing an in-depth analysis of the detector, feature association, and counting experiments. Section 5 discusses the limitations of the StraTracker model and proposes directions for future research. Finally, Section 6 concludes the study.

2. Materials

Data serves as the input to the deep learning model, and the accuracy and precision of the model can be enhanced through a substantial number of correct samples. Currently, the available open-source strawberry datasets are limited in size and application scenarios,

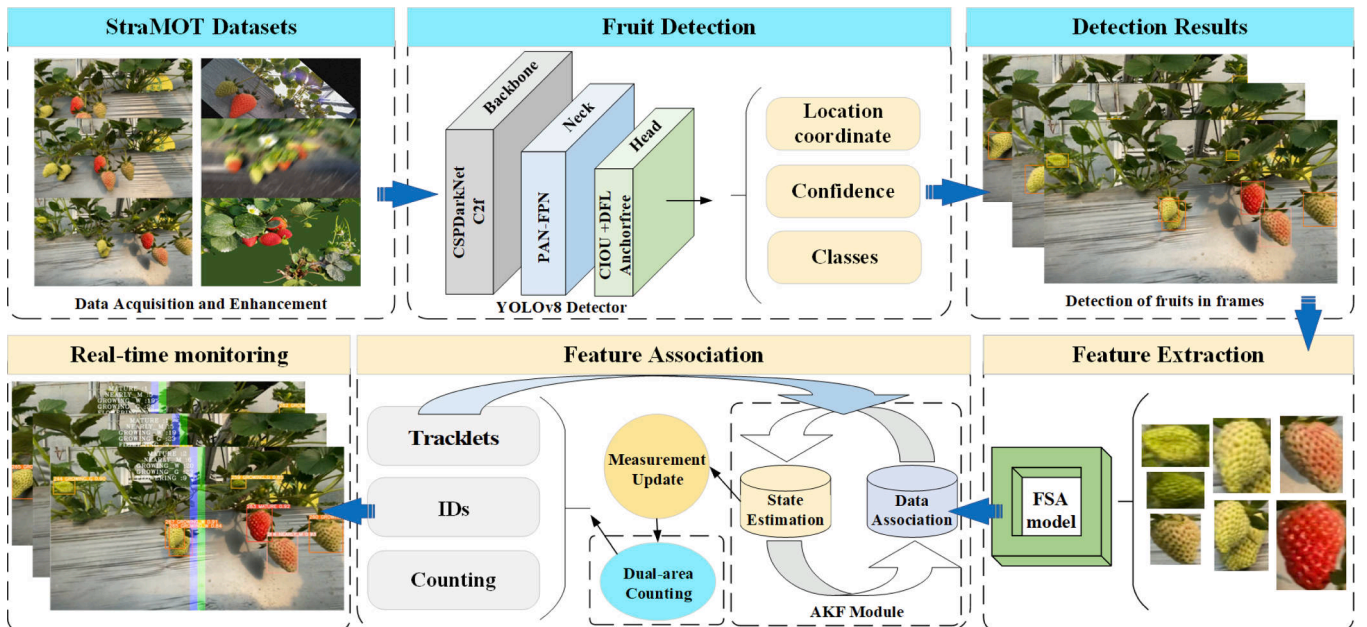


Fig. 1. MOT-based fruit counting architecture.

which provides minimal reference value for our study. Therefore, in this section, we delve into methods for dataset acquisition, growth stages calibration, and data enhancement. Furthermore, we have meticulously curated the StramOT strawberry dataset, specifically designed for the MOT algorithm. This dataset is an invaluable resource that aids in both model training and evaluation.

2.1. Data acquisition

First, we established a strawberry monitoring platform, as shown in Fig. 2(a). The platform's motion system is driven and steered by hub motors, while the microcontroller processes signals from a remote control to regulate the platform's movement. Notably, the platform can traverse strawberry ridges at a speed of 3.7 km/h without causing harm to the growing environment or the fruits. Secondly, between November 2021 and April 2022, as well as October 2022 and March 2023, we conducted data acquisition utilizing the platform in the strawberry greenhouse situated at the Academy of Agricultural Sciences in Weifang City, Shandong Province, China (longitude 119.21866, latitude 36.819597), as illustrated in Fig. 2(b). On the one hand, to ensure the generality and generalization of the strawberry monitoring model, we selected the varieties 'Rouge', 'Suizhu', and 'Zhangji', whose fruit coloring gradually changes to red at maturity. On the other hand, to ensure the diversity of the dataset, we selected different weather and light conditions by collecting in multiple orchards every other week.

2.2. Data calibration

The accuracy of data calibration directly influences model training. Based on the practical experience of senior farmers and relevant industry standards, we classify the growth of strawberry fruits into five stages: flowering (FLOWING), greening (GROWING_G), white ripening (GROWING_W), coloring (NEARLY_M), and ripening (MATURE). Specific calibrations and corresponding labels are detailed in Table 1. According to industry standards for strawberries in both China (NY/T 444-2001) and American (USA Strawberry Grade Standard), strawberries are considered mature and suitable for picking, sale, and consumption when their overall color exceeds 70 %. The flowering period is primarily determined through the detection of strawberry flowers. These flowers typically form cymes featuring white petals that are nearly round or obovate-elliptic. During the white ripening and green fruit stage, strawberries undergo an expansion period where the overall growth transitions from flowers to green fruits. Subsequently, the fruits gradually shift in color, evolving from green to white, and the texture transforms from hardness to a gradual softening.

The dataset is labeled to obtain information about the categories, coordinates, and labels present in each image in preparation for

Table 1
Calibration of strawberry fruit stages.

Number	Fruits	Stages	Labels	Features
0		Mature	MATURE	The fruit's surface color changes to red or deep red and the overall coloring of the fruit is $\geq 70\%$.
1		Coloring	NEARLY_M	The fruits gradually turn red, the surface is partially colored, and the coloring area is $\leq 70\%$.
2		White ripening	GROWING_W	The fruit's surface color gradually changes from green to light white, and the fruits are enlarged in size, and fuller in shape.
3		Greening	GROWING_G	The fruits grow from the stamens into fruits that are green in color on the surface, which are smaller and less full in shape.
4		Flowering	FLOWERING	Cymes with white, suborbicular, or obovate-elliptic petals.

subsequent training of the model. The captured videos were annotated using CVAT, a powerful online image annotation system. The smallest rectangular annotation box encompassing the strawberry is selected as the accurate bounding box(B-box), aiming to minimize the inclusion of background pixels in the actual box. The captured data was stored at a resolution of 1920x1080, with a video frame rate of 60 fps(frames per second) in MP4 format. The specific steps of data labeling are outlined below:

- Sliced the videos in the training subset every 60 frames using Python, resulting in a total of 5,198 valid data frames. Of these, 4,103 frames

Table 2
Details of the StramOT datasets.

Dataset	Video	Average length(min)	Original		Data enhanced	
			Frame	B-box	Frame	B-box
Complete dataset	57	10.46	35,792	—	56,584	—
Train Subset	13	—	4103	27,741	16,412	110,964
Test Subset	—	—	1095	7404	4380	29,616
Counting TestSets	44	—	30,594	—	—	—

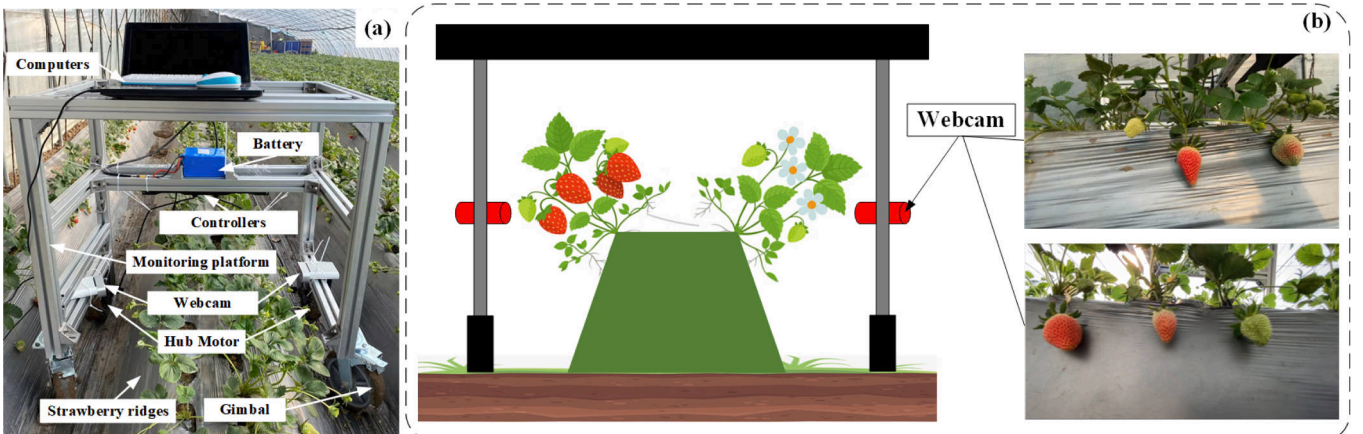


Fig. 2. Monitoring platforms and acquisition methods.

were included in the training subset, while 1,095 frames were assigned to the test subset (see Table 2 for more details).

- Obtained information through the CVAT annotation system, containing Frame ID, object trajectory ID, B-box location, object width and height, classes, and visibility ratio.
- Collected a total of 35,145 strawberry annotation target frames, including those labeled 'MATURE': 6,373, 'NEARLY_M': 7,271, 'GROWING_W': 8,923, 'GROWING_G': 6,941 and 'FLOWERING': 5637 (see Fig. 3 for more details).

2.3. Dataset augmentation

We investigated three hybrid data enhancement methods to augment the original dataset, including RandAugment, motion blur, and AI-for-background, to increase the original dataset to 20,792 valid data, as shown in Fig. 4. The data enhancement is performed as follows:

- RandAugment: randomized augmentation of data by mixing pretzels, amplitude incremental cropping and rotation.
- Motion blur: Combine data randomly mixed linear motion blur, Gaussian blur, mean blur, and median blur in the horizontal direction for enhancement.
- AI changes background: batch the data through AI to remove the background and then randomly generate different backgrounds.

3. Methodology

In this section, we detail the components of the StraTracker model, as shown in algorithm 1. The framework consists of three main parts: the strawberry detector, the feature association module, and the Dual-area Counting (DC) module. In the detection module, the YOLOv8n algorithm was developed to detect strawberries at different growth stages. The feature association module incorporates the Feature Slice Attention (FSA) module, the Adaptive Kalman Filter (AKF) module, and deep fruit feature fusion. The counting module employs the DC method based on the tracked target frames and ID. The workflow of the model is as follows: firstly, the consecutive frames are inputted into YOLOv8 to detect the strawberries and obtain the confidence, position, and category information of the fruits. Secondly, the results are inputted into the feature correlation module to be processed, such as FSA, AKF module, Hungarian Algorithm, etc. Thirdly, based on the tracking results, the DC Module is utilized to realize the counting of the fruits in different growing states by judging the moving direction, ID, and category of the fruits.

Algorithm 1. Pseudo-code of StraTracker

Input: Real-time camera or video sequence V ; Detector D ; Deep fruit feature extractor Enc ; High detection score threshold τ ; New track score threshold η ; Adaptive kalman Filter Module AKF ;
Output: Tracks $\mathcal{T}$ of the V ; Number of strawberries in different growing status S

```

1 Initialization:  $\mathcal{T} \leftarrow \emptyset$ 
2 for frame  $f_i$  in  $V$  do
  /* Update detection results */
3   $D_i \leftarrow D(f_i)$ ;  $D_{high} \leftarrow \emptyset$ ;  $D_{low} \leftarrow \emptyset$ ;  $\mathcal{T}_i \leftarrow \emptyset$ 
4  for  $d$  in  $D_i$  do
5    if  $d.score > \tau$  then
6      /* High score detection storage */
7       $D_{high} \leftarrow D_{high} \cup \{d\}$ 
8      /* Based on the ISA block refinement features */
9       $e \leftarrow FSA(d.box)$  /* See subsection 3.2.1 for details. */
10      $\mathcal{T}_i \leftarrow \mathcal{T}_i \cup Enc(f_i, e)$ 
11   else
12     /* Low score detection storage */
13      $D_{low} \leftarrow D_{low} \cup \{d\}$ 
14     /* Update the prediction trajectory */
15   for  $\ell$  in  $\mathcal{T}$  do
16      $\mathcal{T} \leftarrow CMC(\ell)$ 
17      $\mathcal{T} \leftarrow AKF(\ell)$  /* See subsection 3.2.2 for details. */
18   /* First association */
19    $A_t \leftarrow IoU \& ReID Dist(D_{high}, T)$ 
20    $D_{remain} \leftarrow$  remaining object boxes from  $D_{high}$  and  $D_{low}$ 
21    $\mathcal{T}_{remain} \leftarrow$  remaining tracks from  $\mathcal{T}$ 
22   /* Second association */
23    $A_t \leftarrow IoU Dist(D_{remain}, \mathcal{T}_{remain})$ 
24    $\mathcal{T}_{re-remain} \leftarrow$  remaining tracks from  $\mathcal{T}_{remain}$ 
25   /* Update matching tracks and counting */
26   Update tracklets for AKF and fruit features
27   /* Delete unmatched tracks */
28    $\mathcal{T} \leftarrow \mathcal{T} \setminus \mathcal{T}_{re-remain}$ 
29    $S \leftarrow DC(A_t)$  /* For the counting module, see Algorithm 2. */
30   /* Creating new tracks */
31   for  $d$  in  $D_{remain}$  do
32     if  $d.score > \eta$  then
33        $\mathcal{T} \leftarrow \mathcal{T} \cup \{d\}$ 
34   Return:  $\mathcal{T}, S$ 

```

3.1. Detector

To reduce the inference time and memory usage of the counting model and to improve its detection speed, we integrated the SOTA YOLOv8 algorithm. This algorithm, developed by the Ultralytics team as the latest version of YOLO based on YOLOv5, is primarily utilized for target detection, segmentation, and classification tasks. Moreover, to meet the requirements of lightweight and real-time detection, we opted for the light model YOLOv8n as the detector for StraTracker. As illustrated in Fig. 5, the structure of YOLOv8 utilizes the anchor-free method

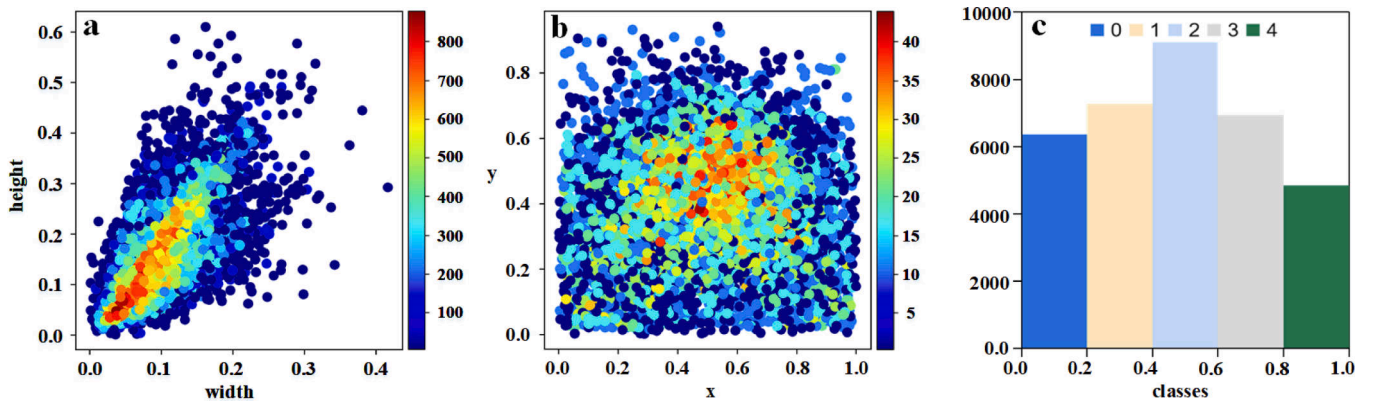


Fig. 3. Illustrates the dataset parameters, where (a) represents the distribution statistics of fruit size, and (b) represents the distribution statistics of center coordinates. The shade of the dot color in (a,b) indicates the density or quantity of data points: blue signifies a low density of data points, while red signifies a high density of data points. Additionally, (c) displays the distribution statistics of the number of five states.

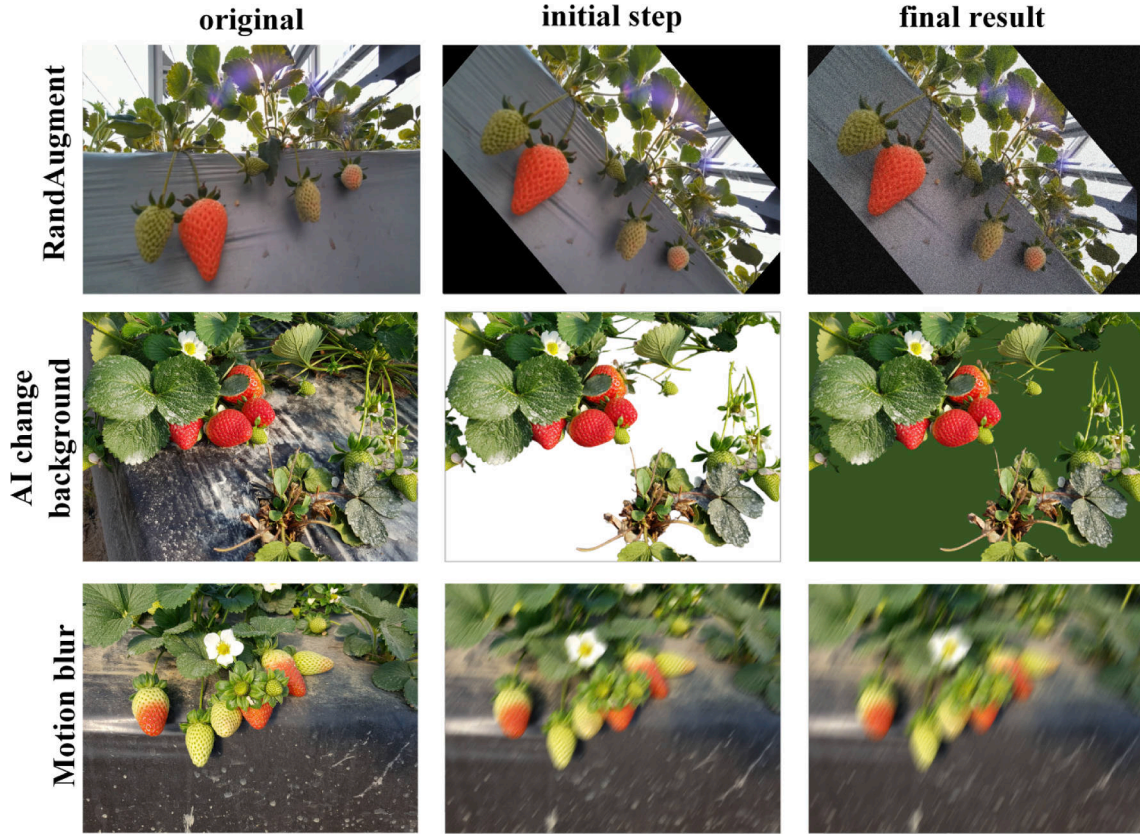


Fig. 4. Dataset enhancement.

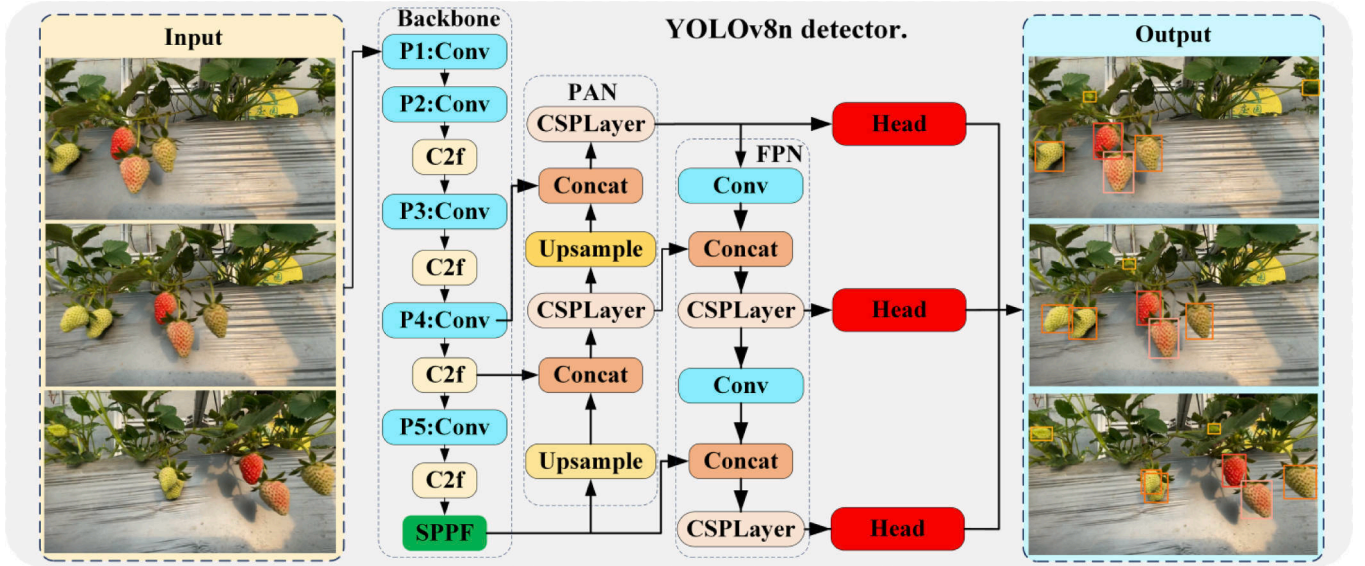


Fig. 5. Structure of the yolov8 algorithm.

based on the original YOLO model. It incorporated the extraction module to the lightweight C2f structure, simplified the PAN-FPN up and down sampling structure, employed the VFL loss as the classification loss, the CIOU Loss + DFL as the regression loss, and adopted the Task-AlignedAssigner sample matching method.

3.2. Feature association

We refer to the BoT-SORT algorithm (Aharon et al., 2022) to design

the association module of StraTracker. Unlike existing MOT algorithms, StraTracker focuses more on fine-grained underlying features due to the small size of strawberry fruits and the presence of light shading. Therefore, we designed the FSA and AKF modules to be incorporated into the association module while retaining the twice feature fusion and deep feature extraction modules.

For more details about the association module, refer to Fig. 6 and Algorithm 1. First, we initialize the trajectory set and traverse the input real-time camera or video sequence frames (V). For each frame, we

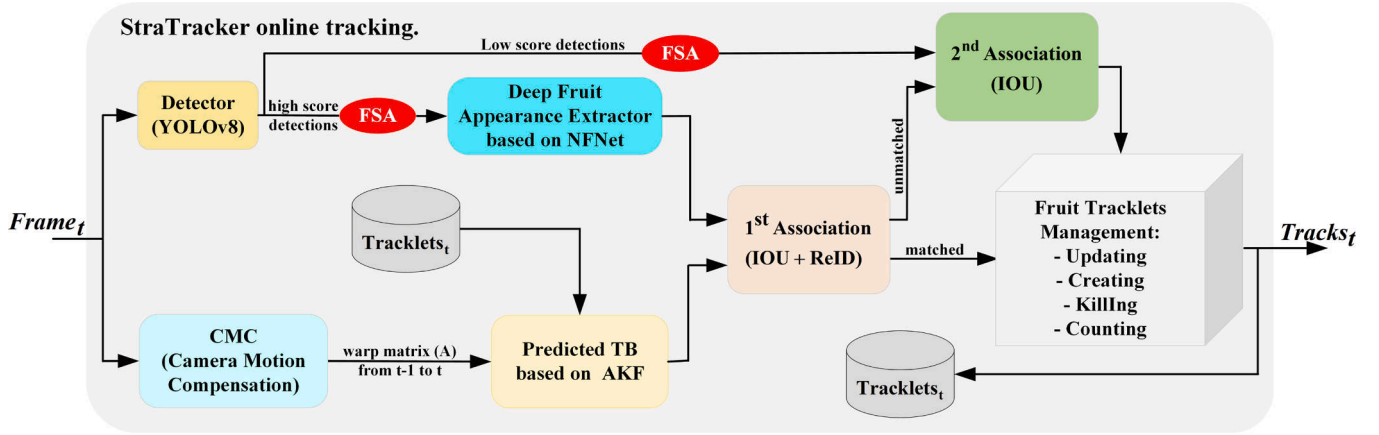


Fig. 6. Feature association framework of StraTracker.

perform the following operations: obtain fruit target information using the YOLOv8 detector and apply a high detection score threshold (τ) to categorize detection into high detection score set (D_{high}) and low detection score set (D_{low}). The fruits in D_{high} are processed through the FSA module to slice their refined features, which are then extracted through the Deep Fruit Feature Extractor (Enc). Meanwhile, the FSA module also improves the accuracy of each frame via Camera Motion Compensation (Cmc) and predicts the fruit target's trajectory ($\mathcal{T}$) using AKF, generating corresponding ID for each trajectory and storing them (line 1–16 in Algorithm 1).

Subsequently, we perform the first feature association using the IOU fusion ReID module, storing the successfully matched tracked targets. We combine the failed matches from D_{high} and D_{low} into D_{remain} . The second association involves matching D_{remain} with the unmatched tracklet trajectories ($\mathcal{T}_{remain}$) using IOU only, and any unmatched target tracking trajectories and detection information are directly deleted.

Finally, the successfully fused tracklet trajectories are continuously monitored and managed, including updating, creating, deleting, and completing the category counting function through the DC module (line 17–25 in Algorithm 1).

3.2.1. Feature slicing attention module (FSA)

During the development of the original BoT-SORT algorithm, variations in fruit size led to challenges, resulting in leakage and false detection problems. Additionally, the existing Reid feature extraction module in the MOT algorithm employed a global search for feature extraction, resulting in prolonged processing times and falling short of real-time tracking requirements. To address these issues, we introduced the FSA module, inspired by the Vision Transformer (Vaswani et al., 2017). The FSA module integrates the target slicing architecture in front of the feature extraction network, enabling the model to rapidly and meticulously extract strawberry appearance features.

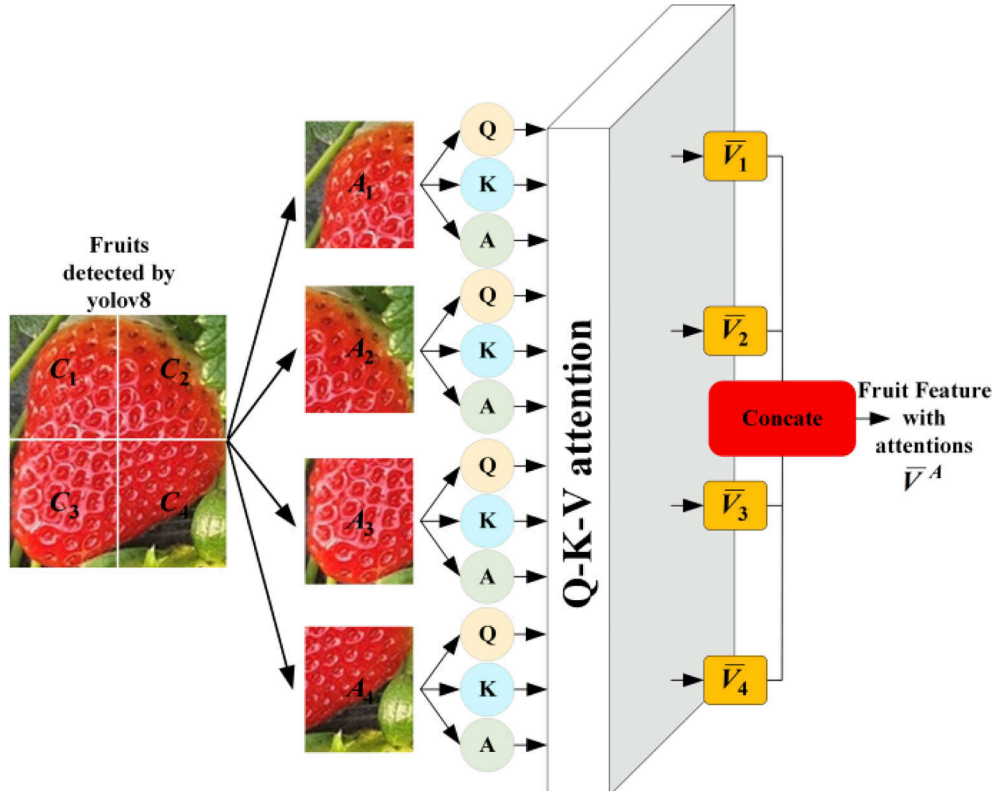


Fig. 7. FSA block.

The FSA block is depicted in Fig. 7. When YOLOv8 provided the acquired fruit coordinates, species, and confidence information to the fruit feature extraction network NFNet, the FSA block performed slicing on each input strawberry feature, dividing it into N individual patches. Subsequently, it embedded all the patches and their corresponding information into three sets of compact features, namely, *QUERY*, *KEY*, and *VALUE*. Leveraging the features from the *QUERY* and *KEY*, we calculated different combinations of attention. The set of values for each patch was then weighted using these attentions, resulting in the formation of a feature vector. Finally, the feature vectors underwent weighting and were incorporated into the feature recognition network. This improvement is intended to increase the accuracy of the model's associations for small target fruits and the speed of monitoring. The core formula of the Q - K - V attention mechanism is depicted in Eq (1).

$$Q - K - V \text{AttenVv} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

3.2.2. Adaptive Kalman filtering module(AKF)

To predict and estimate the motion trajectory of the target on the 2D plane, MOT algorithms universally employ the KF(Mandic et al., 2015) with constant velocity as the prediction module. While the KF can predict the next trajectory of the target accurately during the motion state, it has limitations. The original KF input state matrix is modeled with a constant velocity, causing a mismatch in the tracking frame during platform motion, as depicted in Fig. 8.

The initial state matrix is determined by estimating the aspect ratio of the strawberry target to select the tracking box. During the development of the correlation module, we discovered that the accuracy of the bounding box directly affects tracking and counting precision. Even a small deviation of only 1 cm can lead to two major issues: First, closely positioned target bounding boxes may overlap and interfere with each other, resulting in tracking failures and significant IDs problems. Second, there is an increased likelihood of misidentification, causing other objects to be recognized as strawberries, which affects counting

accuracy. Therefore, we redefined the state vector of the KF in Eq. (2) and Eq. (3) to directly estimate the width and height of the tracking frame.

$$\mathbf{x}_k = [\mathbf{x}_c(t), \mathbf{y}_c(t), \omega(t), h(t), \dot{\mathbf{x}}_c(t), \dot{\mathbf{y}}_c(t), \dot{\omega}(t), \dot{h}(t)]^T \quad (2)$$

$$\mathbf{z}_k = [\mathbf{z}_{xc}(t), \mathbf{z}_{yc}(t), \mathbf{z}_\omega(t), \mathbf{z}_h(t)]^T \quad (3)$$

Where, $[\mathbf{x}_c(t), \mathbf{y}_c(t)]$: the center coordinates of the tracking box; $[\omega(t), h(t)]$: the width and height of the tracking box; $[\dot{\mathbf{x}}_c(t), \dot{\mathbf{y}}_c(t), \dot{\omega}(t), \dot{h}(t)]$: is the speed change value of the corresponding parameter; $\mathbf{z}_k$ is the center coordinates and width/height at the time of measurement.

During KF training and application, the model is updated and corrected in real-time using both real and predicted data. Consequently, it is imperative to tailor the measurement noise $R(k)$ and process noise $Q(k)$ settings to the particular scene. Nonetheless, when the camera undergoes movement or when there are changes in light or the environment, the original $R(k)$ and $Q(k)$ parameters fail to adapt to the real-time scene. This results in tracking loss and a decline in accuracy. Therefore, we introduce the AKF module based on the KF. The AKF dynamically adjusts the gain of the associated module by fine-tuning the weights of the sum. The specific implementation process is as follows.

The first is the prediction process, which is performed for the state quantity $\mathbf{x}$ and the accompanying covariance $\mathbf{p}$, as shown in Eq. (4) and (5):

$$\hat{\mathbf{x}}_{(k|k-1)} = \Phi_{(k,k-1)} \hat{\mathbf{x}}_{(k-1)} \quad (4)$$

$$\mathbf{P}_{(k|k-1)} = \Phi_{(k,k-1)} \mathbf{P}_{(k-1)} \hat{\Phi}_{(k,k-1)}^T + \Gamma_{(K,K-1)} \mathbf{Q}_{(k)} \Gamma_{(K,K-1)}^T \quad (5)$$

Where $\mathbf{P}_{(k|k-1)}$ and $\mathbf{P}_{(k-1)}$ denote the prediction error covariance and error covariance; $\Phi_{(k,k-1)}$ is the system state transfer matrix; and $\Gamma_{(K,K-1)}$ is the system noise matrix.

We introduced the moderating update parameter d_k in KF so that $R_{(k)}$ can be dynamically adjusted based on d_k , and $R_{(k)}$ tends to $R_{(k-1)}$ as the

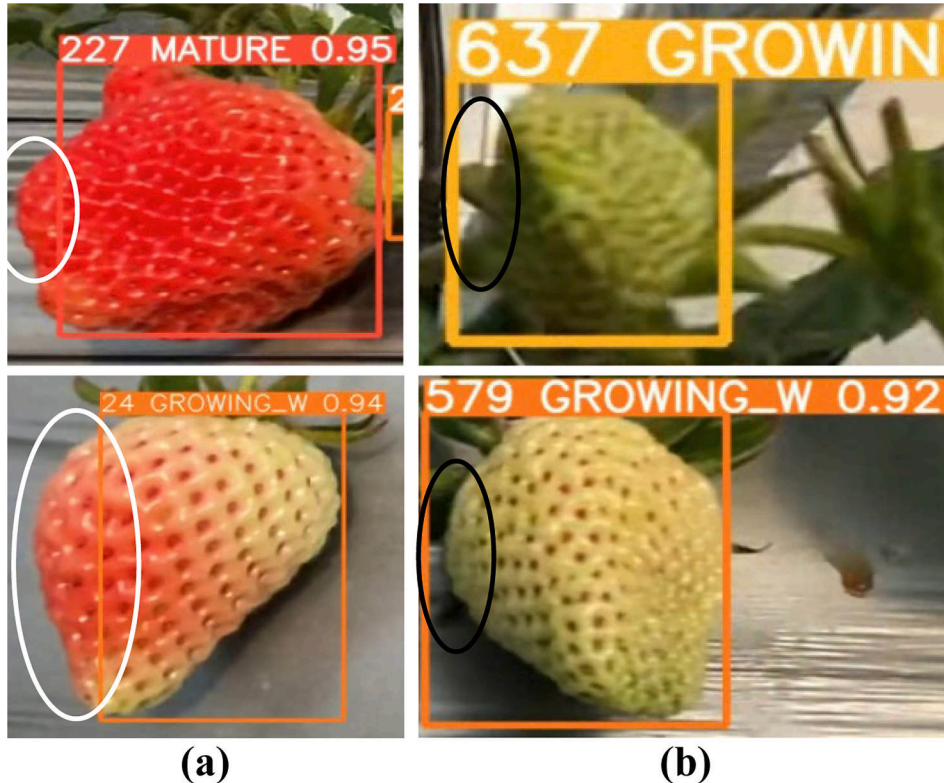


Fig. 8. Problems with the original tracking box, where (a) the tracking frame misses enclosing fruit; (b) the tracking box is too large.

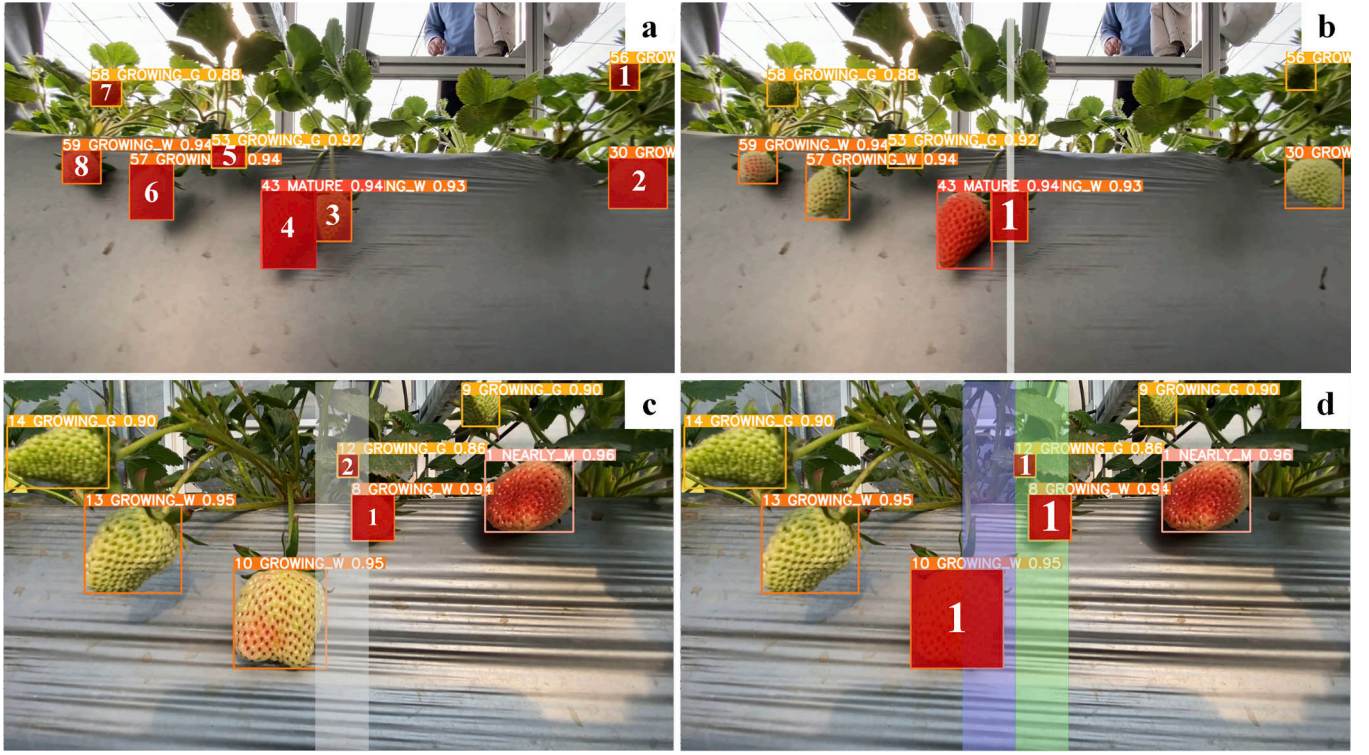


Fig. 9. Comparison of counting methods, where the dark red markers with numbers indicate a successful count.

number of frames k varies, indicating that $R_{(k-1)}$ tends to stabilize, as shown in Eq. (6) and (7):

$$d_k = \frac{1-b}{1-b^{k+1}}, 0 < b < 1 (0.95-0.99) \quad (6)$$

$$v_{(k)} = Z_{(k)} - H_{(k)} \hat{X}_{(k|k-1)} \quad (7)$$

$$Y_k = [I - H_{(k)} K_{(k-1)}] v_{(k)} V_{(k)}^T (I - H_{(k)} K_{(k-1)}) \quad (8)$$

$$R_{(k)} = (1 - d_k) R_{(k-1)} + d_k [Y_k^T + H_{(k)} P_{(k-1)} H_{(k)}^T] \quad (9)$$

where b is the forgetting factor with values ranging from 0.95 to 0.99; $H_{(k)}$ is the system observation matrix; and $v_{(k)}$ is the system observation vector;

$$S_{(k)} = H_{(k)} P_{(k,k-1)} H_{(k)}^T + R_{(k)} \quad (10)$$

The next step is to test the update process, which updates the state vector of the update section:

$$K_{(k-1)} = P_{(k|k-1)} H_{(k)}^T S_{(k)}^{-1} \quad (11)$$

$$P_{(k)} = [I - K_{(k)} H_{(k)}] P_{(k,k-1)} \quad (12)$$

$$\hat{X}_{(k)} = \hat{X}_{(k|k-1)} + K_{(k)} v_{(k)} \quad (13)$$

3.3. Count module

To accurately count strawberries during the growing period, we integrated a cross-frame counting module into StraTracker and explored four counting methods through comparison. The specific counting methods are as follows:

1. ID-based counting(IC): An ID will be assigned to each target during the monitoring process, and strawberry fruits with different growth

stages will be counted based on ID, respectively (X. Li et al., 2022), as illustrated in Fig. 9(a). This method heavily depends on the performance of the tracking and detection algorithm, and the detection results often exceed the true value.

2. Line-based counting(LC): A counting line is incorporated into the algorithm to indicate the center coordinates of the tracking box and estimate when the trajectory of the target center intersects with the counting line (S. Li et al., 2020), as shown in Fig. 9(b). This method is prone to a significant number of missed detections due to the limited range of counting lines.
3. Area range-based counting(AC): Integrate a region range into the algorithm to detect the target's tracking box as a signal and count the strawberry tracking box when it is detected within the region range, as shown in Fig. 9(c). This method has the potential to reduce the tracking range and enhance the counting accuracy of the model. However, it is limited in its ability to categorize strawberry targets, only providing a total count.
4. Based on dual-area counting(DC): Introduce two judgment regions into the algorithm to monitor the tracking box of the fruit, as shown in Fig. 9(d). When the tracking box passes through the first region, confidence judgment is performed, and its corresponding ID is stored. Upon reaching the second region, the growth stages is assessed, and the ID is reevaluated for classification in the count. Refer to Algorithm 2 for the pseudo-code of our DC method (line 1–10 in Algorithm 2).

We conducted several experiments on these four fruit counting methods, as described in Section 4.4. The results indicate that the DC method is the most efficient and capable of monitoring the overall growth of fruit in the orchard.

Algorithm 2. (Pseudo-code of Dual-area Counting (DC))

Input: Fusion matrix A_i (Includes d and $\mathcal{F}$); Two judgment regions $R1$ and $R2$; High detection score threshold τ

Output: Number of strawberries in different growing status S

(continued on next page)

(continued)

```

1 Initialization:  $S \leftarrow \{ \text{FLOWING: 0, GROWING\_G: 0, GROWING\_W: 0, NEARLY\_M: 0, MATURE: 0} \}; R1\_IDs \leftarrow \emptyset$ 
2 for  $t$  in  $A_t$  do
    /* Confidence judgment in R1 */
3   If  $d.\text{box}$  through R1 then
4     If  $d.\text{conf} > \tau$  then
5        $R1\_IDs \leftarrow R1\_IDs \cup \{ \mathcal{T}.ID \}$ 
6       store  $R1\_IDs$ 
    /* Assess growth status in R2 */
7   If  $d.\text{box}$  passes R2 then
8     If  $\mathcal{T}.ID$  in  $R1\_IDs$  then
9        $S\{\text{growth\_status}\} + = 1$ 
10 Return:  $S$ 

```

3.4. Evaluation metrics

We selected *HOTA* (Higher Order Tracking Accuracy), *MOTA* (Multi-Object Tracking Accuracy), *MOTP* (Multi-Objective Object Tracking Precision), *mAP* (Mean Detection Precision), and *IDs* as the evaluation metrics for strawberry tracking models. Where the *IDs* represent the number of IDs, which is the change of predicted trajectory ID for the same target GT trajectory between consecutive frames. Ideally, the ID should remain constant, but various factors can cause the model to generate multiple ID for the same target, thus affecting counting accuracy. The larger the number of IDs, the worse the model performance.

HOTA encompasses higher latitude accuracy metrics, providing a more balanced and comprehensive evaluation of the model's tracking performance.

$$HOTA = \sqrt{\frac{\sum_{c \in \{TP\}} A(c)}{TP + FN + FP}} \quad (14)$$

$$A(c) = \frac{TPA(c)}{TPA(c) + FNA(c) + FPA(c)} \quad (15)$$

Where $A(c)$ denotes association accuracy, $TPA(c)$ denotes accurate association accuracy; $FNA(c)$ denotes accuracy of unassociated predicted trajectories, $FPA(c)$ indicates the accuracy of predicting incorrectly associated trajectories, FP denotes the number of misdetected targets, FN denotes the number of lost targets, and TP denotes the number of targets used for correct detection of positive samples.

MOTA denotes Multi-Objective Monitoring Accuracy, which visualizes the monitoring accuracy of the model by combining the error rates of False Detection, Missed Detection, and IDs.

$$MOTA = 1 - \frac{\sum_t (FP + FN + IDs)}{\sum_t g_t} \quad (16)$$

Where t denotes the t -th frame, g_t represents the number of all correct targets, and *IDs* indicate the number of target ID switches per frame.

We also refer to the metrics designed in the counting algorithm by X. Li et al. (2022) to evaluate the model's counting results. The counting metrics are *Accuracy* and *GEH*, as shown in Eq (17) and Eq (18). *Accuracy* denotes the accuracy of the counting results, and *GEH* is a classical quality metric for verifying counting accuracy.

$$Accuracy = 1 - \frac{Ds}{G_t} \quad (17)$$

$$GEH = \sqrt{\frac{2 \times Ds}{C_t + G_t}} \quad (18)$$

Where G_t denotes the true number of fruits, C_t denotes the model count results, and *Ds* represents the deviation of the real situation from the count results.

4. Experiments

Several sets of experiments were conducted to validate the efficacy and utility of StraTracker. (1) detector experiments: Experiments were carried out on different lightweight deep learning detectors, involving a trade-off analysis between real-time processing and detection accuracy. The goal was to identify the most suitable detector for integration into StraTracker. (2) feature association experiments: We compared the proposed StraTracker with a variety of popular MOT algorithms and conducted ablation experiments and counting tests. (3) field dynamic experiments: dynamic evaluation of strawberries was conducted in the orchard based on the experimental monitoring platform, and real-time strawberry growth distribution and occupancy maps were generated.

4.1. Experimental environment

In this study, the StraTracker model underwent training on a workstation equipped with an NVIDIA RTX 3090 GPU, coupled with an Intel (R) Xeon(R) Gold 6234 @3.30 GHz processor and 192 GB of RAM. The programming language employed was Python 3.8, and the deep learning framework utilized was PyTorch 1.7.1 GPU version. Subsequent tests were conducted using Intel's NUC minicomputer, featuring an NVIDIA GeForce RTX 2060 GPU, 32 GB of RAM, and an Intel(R) Core(TM) i7-1165G7 @2.80 GHz processor.

4.2. Detector experiments

Detectors are essential for accurately identifying the growth stages and ripeness of strawberry fruits. To determine the most effective model, we evaluated several lightweight detection algorithms, including Fast-RCNN, RetinaNet, NanoDet, YOLOv5s, and YOLOv8n, using the StraMOT dataset. The experimental results, detailed in Table 3 and Fig. 10, revealed distinct performance differences among these models. Fast-RCNN, due to its high parameter count and two-phase structure, exhibited poor accuracy and real-time performance, achieving a mAP of only 73.26 %. In contrast, YOLOv5s demonstrated higher real-time FPS but slightly lower detection accuracy, particularly in the colouring phase, where it achieved an accuracy of only 88.86 %. NanoDet, while being a lightweight model, also showed lower accuracy and unsatisfactory real-time performance for detecting strawberry growth stages. Among the tested models, YOLOv8n emerged as the best-performing lightweight detector, with a mAP of 93.74 % for ripening fruits, an F1 score of 90.46 %, and a minimal parameter count of just 3.2 M. This made YOLOv8n the most suitable model for monitoring the growth and ripeness of strawberries.

Notably, key parameters, including the real-time loss function, mAP, F1 score, and FPS, were visualized throughout the training process of the five detection algorithms, as depicted in Fig. 10. This visualization provides an intuitive depiction of each detector's strengths and weaknesses. In particular, the similarity between the training and validation loss curves suggests that the model is not overfitting. Overfitting, which occurs when a model performs well on training data but poorly on unseen data, was effectively mitigated by optimizing the dataset and applying a regularization strategy to control model complexity. Moreover, YOLOv8n demonstrates a tendency toward stabilization in its loss function after 40 epochs, indicating faster convergence, while also achieving an impressive mAP of 91.93 %, signifying superior detection accuracy. Additionally, on the StraMOT dataset, YOLOv8n attains a real-time monitoring frame rate of 38.3 fps, demonstrating effective real-time detection capabilities. Thus, we have chosen YOLOv8n as the detector for the StraTracker model.

4.3. Feature association experiments

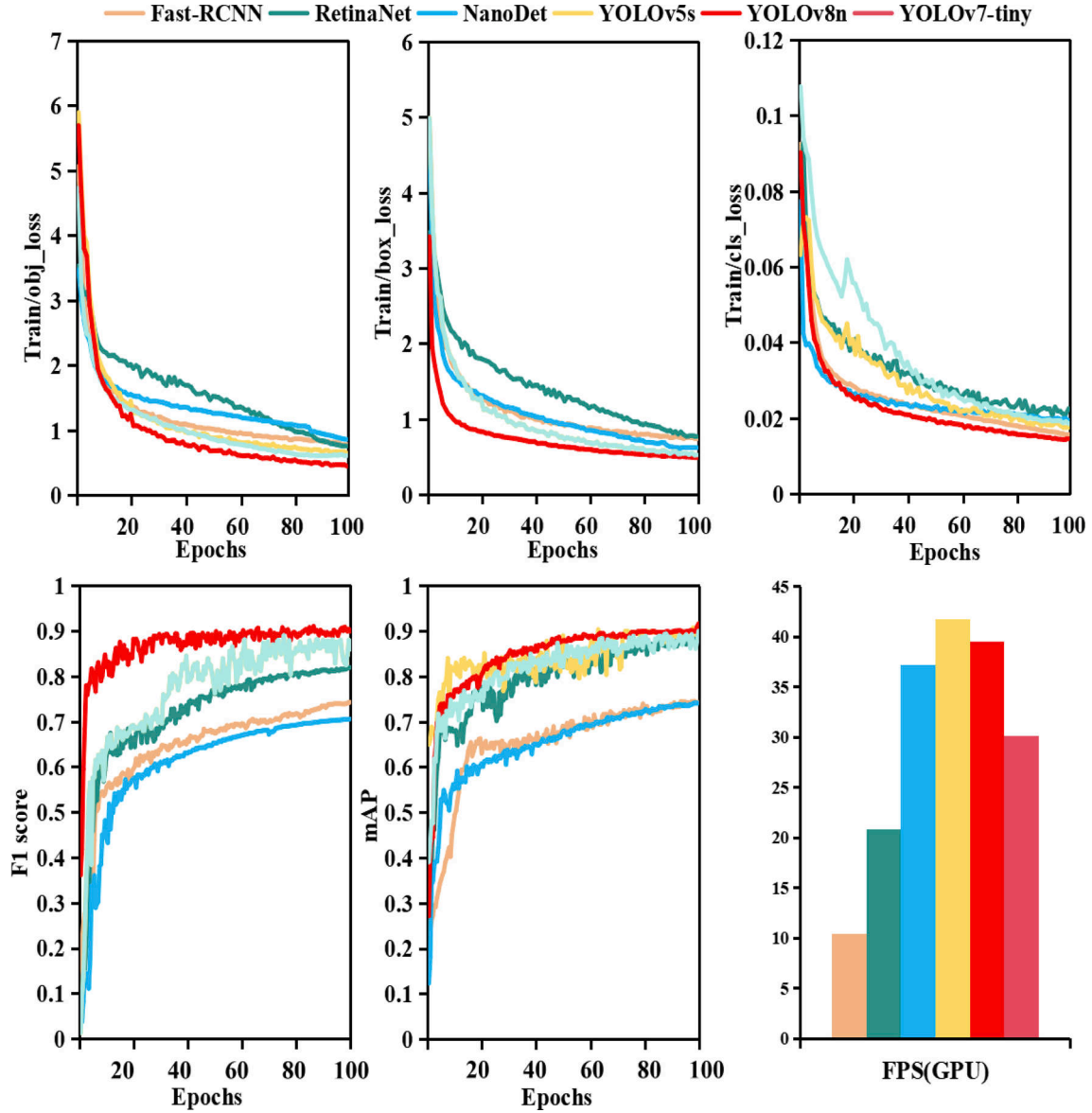
4.3.1. Ablation experiment

Ablation experiments were conducted on the StraMOT dataset to

Table 3

Comparison of the performance of different detectors.

Detector	AP(%)↑					FPS↑	Params↓
	MATURE	NEARLY_M	GROWING_W	GROWING_G	FLOWERING		
Fast-RCNN	75.85	73.05	71.29	71.83	74.28	10.4	330.3 M
RetinaNet	85.08	87.84	85.32	84.57	87.95	20.8	36.3 M
NanoDet	75.72	74.66	72.03	72.15	75.28	37.2	2.4 M
YOLOv5s	92.47	88.86	90.13	90.60	91.91	41.7	7.2 M
YOLOv7-tiny	92.97	90.16	89.71	89.23	91.49	30.1	6.1 M
YOLOv8n	93.74	90.31	91.63	92.35	91.64	38.3	3.2 M

**Fig. 10.** Experimental comparison of the detector.**Table 4**

Comparison table of ablation trials, where DA indicates data augmentation.

BoT-SORT	DA	YOLOv8	AKF	FSA	MOTA↑ (%)	MOTP↑ (%)	HOTA↑ (%)	IDs↓	FPS↑
✓	—	—	—	—	78.52	80.52	74.39	330	32.7
✓	✓	—	—	—	78.83	80.71	74.85	321	32.5
✓	✓	✓	—	—	81.01	80.94	76.41	294	35.7
✓	✓	✓	✓	—	82.72	81.13	76.93	267	37.2
✓	✓	✓	✓	✓	83.28	81.35	77.26	259	38.3

validate the effectiveness of each proposed module. Table 4 presents the results using BoT-SORT as the baseline, where we tested the performance of the algorithm by adding the following modules: data augmentation (DA), YOLOv8, AKF, and FSA blocks. The experimental outcomes show that the detector significantly enhances the model's tracking performance. Specifically, integrating the YOLOv8n algorithm improved MOTA, MOTP, and HOTA by 2.18 %, 0.23 %, and 1.56 %, respectively, representing a substantial gain. The AKF module had the most significant impact on the IDs metric, reducing it by 33. The FSA module further enhanced the accuracy of tracking small target fruits, raising MOTA to 83.28 % and HOTA to 77.26 %. DA also contributed to improved model accuracy and reduced overfitting. Overall, the model achieved a 4.76 % improvement in MOTA, a 0.83 % increase in HOTA, and a 2.87 % gain in MOTP. Furthermore, fps increased from 32.7 to 38.3, while IDs decreased from 330 to 259, demonstrating notable improvements.

The tracking effects of the original ground truth (GT), BoT-SORT, and StraTracker models are compared in Fig. 11. Fig. 11(a) displays the GT, where boxes with the same color labels correspond to the same fruit. In Fig. 11(b), the labeled white boxes indicate instances where the BoT-SORT algorithm encounters problem such as misfitting tracking boxes and missed detections. Additionally, the model exhibits problems with low confidence and IDs. In Fig. 11(c), the accuracy of the tracking boxes for fruit selection is superior, attributed to the modification of the input state matrix in the AKF module of the StraTracker algorithm. The StraTracker model accurately recognized the growth of the strawberry and assigned specific IDs from frame 29 to frame 38. Throughout the process, it successfully detected small target fruits without any IDs or omissions. This performance is attributed to the integration of the FSA and AKF modules, which significantly enhance the model's adaptability and ability to monitor small targets.

4.3.2. Comparison of the different models

To validate the superiority of the StraTracker model, we conducted cross-vertical comparison experiments using the StraMOT dataset described in Section 2. The tracking performance experiments evaluated various MOT algorithms and the original model with modified YOLOv8

detectors. Specifically, we selected six widely recognized MOT algorithms for comparison: SORT(Bewley et al., 2016), DeepSORT(Pujara & Bhamare, 2022), ByteTrack(Y. Zhang et al., 2022), OCSORT(Cao et al., 2023), StrongSORT(Du et al., 2023), and BoT-SORT(Aharon et al., 2022). Unlike existing methods, these algorithms required modification and training to track and monitor strawberries during their growing stage effectively. Additionally, each model utilized pre-training weights obtained from training on the MOT17 dataset and underwent training on the StraMOT dataset for 100 epochs.

The results of the experiments are presented in Table 5, where the StraTracker model outperformed the other methods across all metrics, achieving a MOTA value of 83.28 % with only 259 occurrences of IDs. In contrast, the SORT and DeepSORT models exhibited lower multi-target accuracy. This discrepancy was likely due to their use of cascade matching in the feature association module, which was more suitable for tracking large targets such as pedestrians and vehicles, rather than small

Table 5

Cross-vertical comparison experiment of MOT algorithms. Where $\uparrow$ indicates a positive correlation, signifying that a larger value corresponds to better model performance; $\downarrow$ indicates a negative correlation, suggesting that a smaller value corresponds to better model performance.

Tracker	MOTA $\uparrow$ (%)	MOTP $\uparrow$ (%)	HOTA $\uparrow$ (%)	IDs $\downarrow$	FPS $\uparrow$
SORT	63.32	66.61	55.85	682	9.3
DeepSORT	70.44	77.69	58.36	514	18.2
ByteTrack	77.83	79.56	68.72	361	38.6
OCSORT	76.18	78.41	73.16	397	43.5
BoT-SORT	78.52	80.52	74.39	330	32.7
StrongSort	81.69	79.05	73.83	318	30.9
DeepSORT (YOLOv8)	72.15	78.39	60.05	437	22.5
OCSORT(YOLOv8)	76.82	79.13	73.92	383	43.9
StrongSort (YOLOv8)	81.75	79.63	74.35	311	31.8
BoT-SORT (YOLOv8)	80.57	80.81	74.89	305	34.2
StraTracker(our)	83.28	81.35	77.26	259	38.3

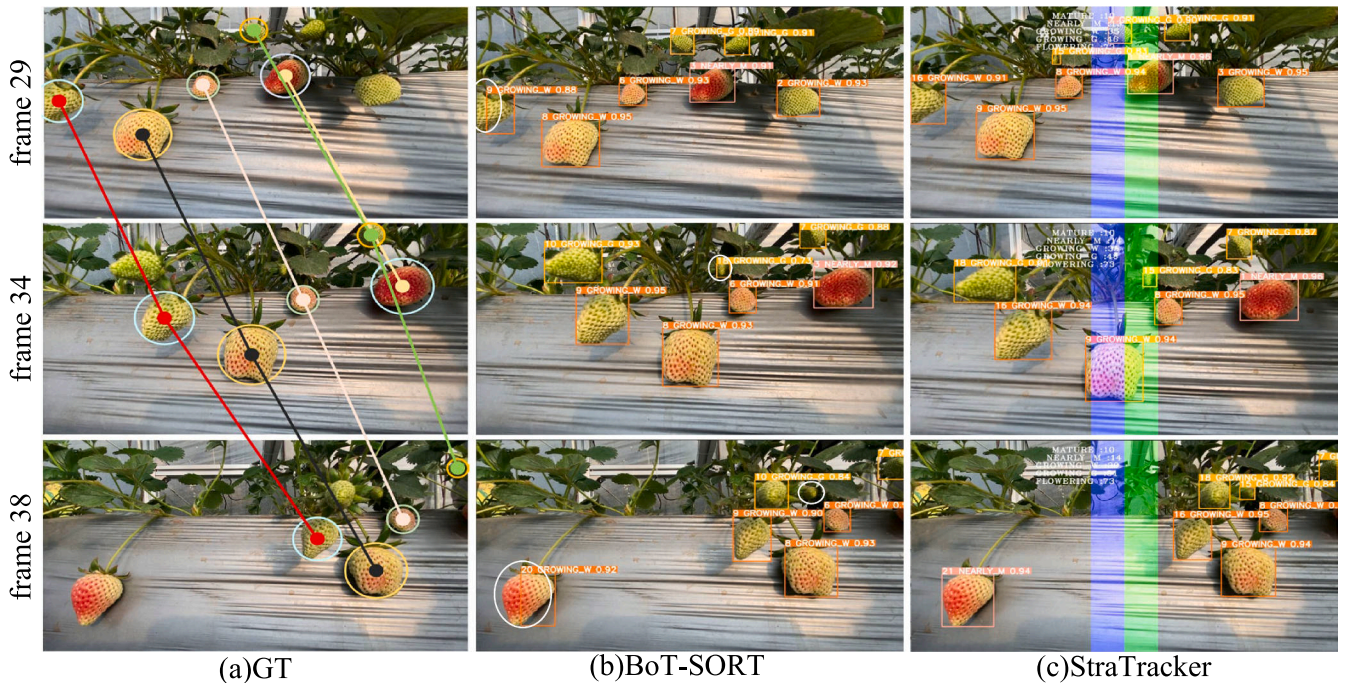


Fig. 11. Comparison of tracking effect before and after improvement. The same color circles in Fig. (a) indicate the same fruits in different frames; the white circles in Fig. (b) indicate the occurrence of box prediction frames that are not fit and IDs problems.

objects like strawberries. Despite recent advances in MOT algorithms, models such as OCSORT, ByteTrack, and StrongSort showed limited effectiveness in tracking and monitoring strawberry growth in natural environments. For instance, OCSORT introduced a motion model and simplified the detector, achieving the best real-time performance at 43.9 fps. However, its MOTA was only 76.18 %, with 397 IDs. StrongSort, which utilized the NAS nonlinear KF module to predict the state of targets with more complex motion patterns, showed poorer real-time performance, reflected in a HOTA of 73.83 %. Through data-level analysis, the StraTracker model demonstrates optimal feature association ability, attributed to the FSA and AKF modules designed to enhance the MOT model's suitability for agricultural applications.

Improving the detector of the MOT model to YOLOv8 enhanced the accuracy of all the algorithms, with the most significant improvement seen in the BoT-SORT algorithm, which is the method used by

StraTracker. This proves the effectiveness of our proposed model. In terms of real-time performance, our designed model achieved 38.3 fps, an improvement of 6 fps over the BoT-SORT algorithm. However, it still lags behind ByteTrack and OCSORT, primarily due to the inclusion of a DC module, which affects its real-time efficiency. Nevertheless, when combining the evaluation metrics, the StraTracker model demonstrated a clear overall advantage.

In Fig. 12, we evaluated the model's effectiveness, where white boxes denote missed or misdetected targets. Particularly in Fig. 12(a), it is evident that the Deepsort model encounters a more severe problem of missed and misdetected targets, directly affecting counting accuracy. Additionally, in Fig. 12(c, d), the ByteTrack and OCSORT models display issues related to tracking frame mismatch and IDs. Overall, these models struggle to effectively handle challenges such as high occlusion, overlap, and IDs.

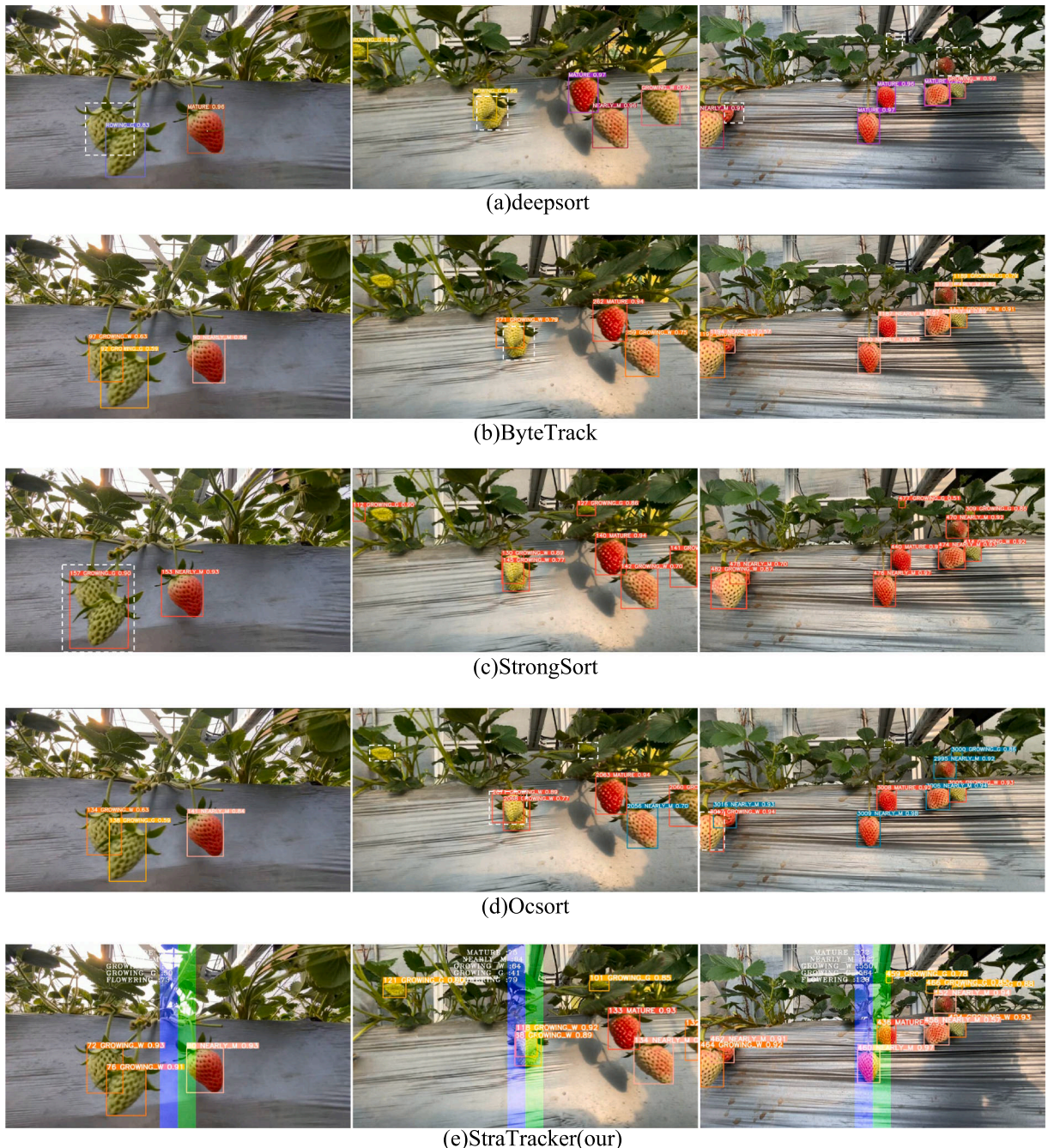


Fig. 12. Comparison of different algorithms. The white dashed boxes in the figure indicate missed or incorrectly detected targets.

However, the StraTracker model was designed with the aim of solving the challenge of monitoring and counting strawberry fruits at different growth stages in natural environments. As shown in Fig. 12(a, b, c, d), all models suffer from leakage and false detection problems. In contrast, the StraTracker model tracks accurately. We believe that the design of the FSA and AKF modules enables the StraTracker model to quickly and meticulously extract the appearance features of strawberries and dynamically adjust the gain of the relevant modules, thereby improving the model's tracking performance on highly occluded and overlapping targets. As a result, the StraTracker model shows excellent performance in monitoring strawberry growth stages.

4.4. Counting experiments

4.4.1. Experimentation with different methods

The counting module directly affects the usefulness of the model. We compared the proposed counting modules to select the most suitable counting method for StraTracker. Four counting methods were employed: IC (ID counting), LC (line counting), AC (area counting), and DC (dual-area counting). The results are shown in Table 6, where the DC method proves to be the most effective, achieving an accuracy of 90.43 %. The GEH is only 2.33, and The Coefficient of determination(R^2) is as high as 0.91. However, due to the DC method being designed with the ID and growth stages judgment function, it results in an increased model detection time, leading to a slightly smaller FPS compared to several other methods, with only 38.3 fps. however, the method counts all IDs in the frame, including ID conversions and false detection targets. This results in the reported count being much larger than the true value. Our experiments also indicated that the counting range is most effective between 1/5 and 1/10, with the best balance observed at 1/8. At this setting, the model maintains high accuracy and achieves reasonable real-time processing efficiency. While the IC method offers a higher frame rate, it does not match the accuracy of the DC method.

Fig. 13 illustrates the linear regression between the predicted and actual quantities obtained by applying the four counting methods in the test data. We extracted 44 videos from the StraMOT dataset as the sample. The X-axis represents the actual number of manual counts, and the Y-axis represents the estimated number. Each point on the plot represents one sample data, and the straight line represents the fitted linear regression equation. From Fig. 13(a), it was observed that the overall IC method detected a large number of poor counts with an R^2 of only 0.68. In Fig. 13(b) and (c), using LC and AC methods, respectively, the accuracy is 87.72 % and 83.15 %. We consider that the LC method's collision line detection range is too small, leading to a significant number of missed detections. On the other hand, the AC method could not determine the growth stages of the fruits and could only obtain the overall number of fruits. Fig. 13(d) displays the linear regression of the DC method with the equation of $Y = 0.9841X - 0.4501$ and the R^2 of 0.91, suggesting optimal performance for the counting method. Therefore, the DC method is more suitable for application to the StraTracker model.

4.4.2. Experiments with different conditions

During the data collection process, we discovered that varying lighting conditions and webcam angles significantly impact the accuracy of the model's counting. Consequently, we evaluated the model's performance under actual orchard conditions and tested its counting

function using the StraMOT dataset. The results are detailed in Table 7 and Fig. 14.

First, we primarily considered sunlighting, backlighting, and occluded light under various lighting conditions, as illustrated in Fig. 14 (a-c). Under backlighting conditions, exposure and background darkening occur, impacting the feature association module and counting function. This lead to association failures or missed counts, resulting in an accuracy and GEH of only 86.04 % and 2.87, respectively. In contrast, under both smooth and shaded lighting conditions, the counting results were more accurate, with accuracy rates exceeding 93 %.

Moreover, We evaluated the monitoring effect on fruits at angles of 0°, 30°, and 45°, as depicted in Fig. 14(c-f). The highest counting accuracy was achieved when the camera was positioned directly at 0°, with an accuracy of 93.97 % and a GEH of 0.96. As the angle increased, significant occlusion of fruits and leaves occurred, providing a wider field of view but also introducing redundant information. This increased monitoring complexity, leading to misses and misdetections. Specifically, at 45°, the model's counts exceeded the true value due to the detection of fruits in adjacent rows, resulting in redundant counts.

In summary, enhancing the model's counting accuracy under different lighting conditions, particularly backlighting, is crucial and provides a direction for future research. Preliminary experiments verified that the monitoring model was most effective at a 0° angle under sunlight conditions. Therefore, for optimal results, the camera angle should be controlled between 0° and 30°.

4.5. Field dynamics experiment

Real-time visualization of the distribution of fruits in the orchard can assist growers in performing timely tasks such as flower thinning, pollination, fertilization, or picking. Therefore, we conducted testing of the StraTracker model in the orchard using the monitoring platform from December 2022 to January 2023. We selected four high ridges in the orchard for validation, with the total length of the ridges being approximately 100 m, as illustrated in Fig. 15 for the path planning diagram.

The StraTracker model enables us to accurately obtain the distribution and occupancy of strawberries in the orchard. As depicted in Fig. 16, we visualized the distribution of the five growth stages of strawberries in each ridge through stacked plots and the overall orchard distribution through percentage plots. In the distribution map, a, b, c, and d represent the four high rows in the orchard corresponding to the planning trajectory in Fig. 15. These are visualized in the form of percentages, with different colors representing different growth stages.

In Fig. 16(a) and (d), the white ripening stage in the overall orchard reached a maximum of 33.84 %, while the ripening stage accounted for only 6.79 %. The analysis indicates that the strawberries in the orchard at this time are predominantly in the early stage of planting and development. It is crucial to implement early treatments, such as flower thinning and pollination, for them in a timely manner. By January 2, the proportion of white ripening and coloring stage fruits reached 24.41 % and 27.43 %, respectively. This suggests that the fruits in the orchard are in the process of expansion in anticipation of maturity. To support their development, additional measures such as spraying, fertilizing, watering, and other protective measures are needed. As of January 10, the ripening stage reached 30.64 %, indicating an overall ripening of fruits in the orchard. Timely picking is now necessary. The StraTracker model not only captures the growth of strawberries in the orchard but also predicts the date of fruit ripening, enabling the scheduling of subsequent management tasks in a timely manner.

5. Limitations and future work

All in all, various experiments have demonstrated the ability of the StraTracker model to accurately monitor the growth of strawberries in orchards and count them in real time. However, there are limitations

Table 6
Comparison experiment of counting methods.

Counting method	Accuracy(%)↑	GEH↑	R^2 ↑	FPS↑
IC	80.64	4.85	0.68	43.9
LC	83.15	4.19	0.76	41.3
AC	87.72	3.02	0.87	41.7
DC	90.43	2.33	0.91	38.3

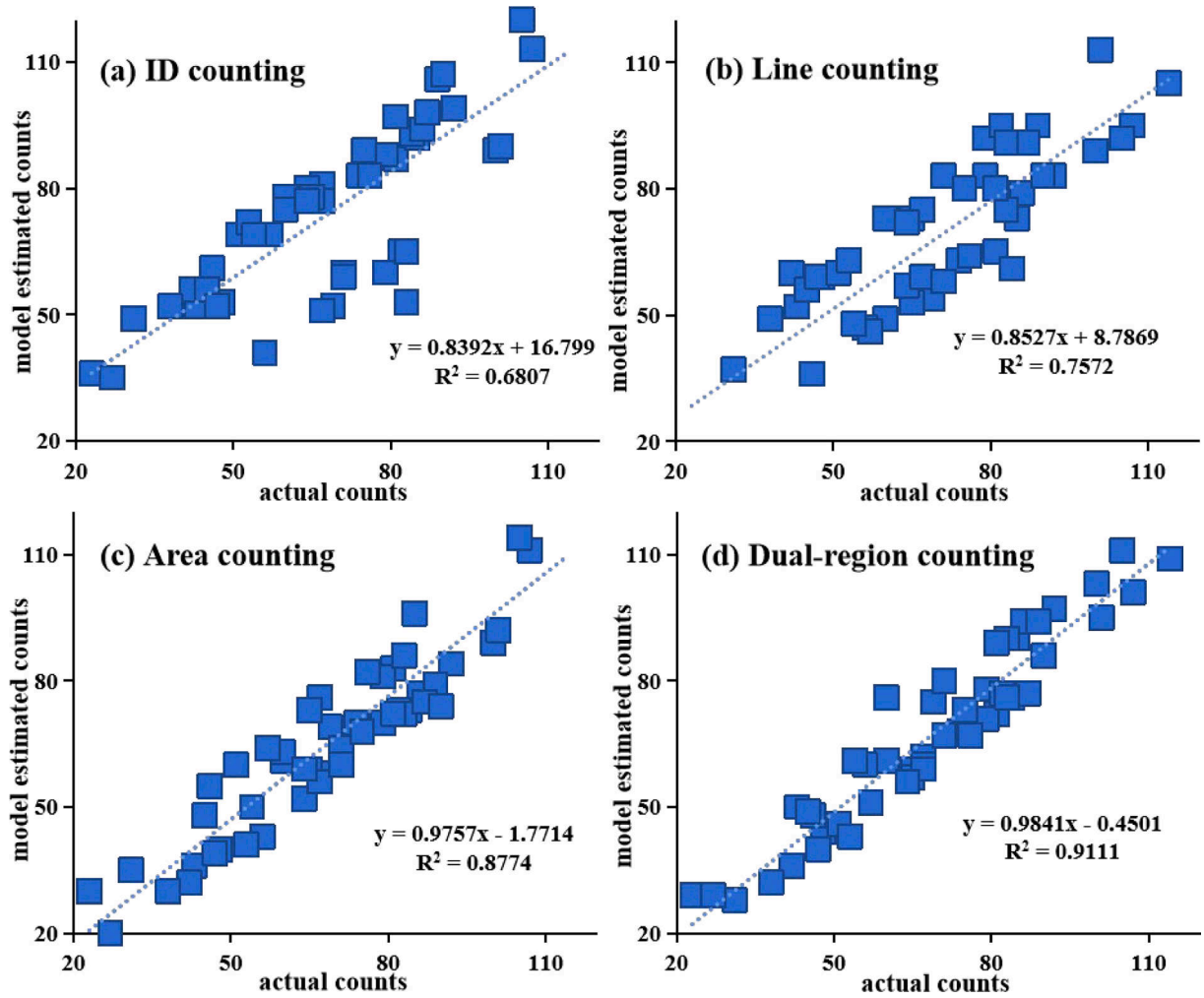


Fig. 13. Comparison of counting regression performance.

Table 7
Comparative experiments on counting conditions.

Different conditions		GT	StraTracker	Accuracy(%) $\uparrow$	GEH $\uparrow$
Shunlight	0°	249	234	93.97	0.96
	30°	192	176	91.61	1.25
	45°	261	287	90.03	1.36
Backlight	0°	394	339	86.04	2.87
	30°	458	393	85.81	3.15
	45°	352	394	83.90	3.52
Occluded light	0°	315	291	92.38	1.37
	30°	157	134	85.35	3.71
	45°	216	238	89.81	2.46

that need to be addressed. Specifically, the model's environmental suitability and counting accuracy require further improvement, and the effects of different factors on counting accuracy need to be investigated experimentally. Future work will focus on enhancing the model's performance in monitoring growth stages, optimizing the counting scheme, and validating the model's applicability to other crops.

To achieve these goals, we will discuss future directions in terms of hardware, software, datasets, and experimental design. In terms of hardware, we will enhance the platform's mobility and stability, and improve the resolution and quality of the camera. For software, we will develop user-friendly interactive interfaces, more accurate data association algorithms, and improved counting schemes. We will also extend the monitoring capabilities to include additional crop characteristics

such as pests, diseases, phenotypic information, and fruit location. Regarding datasets, we will increase the diversity of crop data, integrate information from various environmental scenarios and cultivation methods, and research data enhancement methods to improve robustness. Experimentally, we will explore the effects of platform movement speed, different cultivation methods, and various crop varieties on the model's monitoring and counting functions.

6. Conclusions

Real-time, accurate, and efficient identification and counting of strawberry growth are crucial for the orchard management industry. In this paper, we propose a novel strawberry tracking algorithm (StraTracker), specifically designed for identifying and counting strawberry growth stages. Firstly, we developed a large-scale strawberry tracking dataset (StraMOT), for training and testing the proposed model. Secondly, we selected the lightweight YOLOv8n detector to reduce inference time and improve the speed of fruit growth stage detection. Furthermore, we introduced the AFK and FSA modules in feature association to enhance small target tracking and optimize adaptability. Finally, we designed the DC module, a fruit counting method that tracks real-time monitored fruits based on detected growth stages and IDs using a dual-area approach. Experimental results demonstrate that StraTracker achieves MOT, MOTP, and accuracy scores of 83.28 %, 81.35 %, and 90.43 %, respectively, with only 259 IDs. These outcomes surpass existing baseline models while meeting real-time processing requirements, with a speed of 38.3 frames per second. Moreover, real-

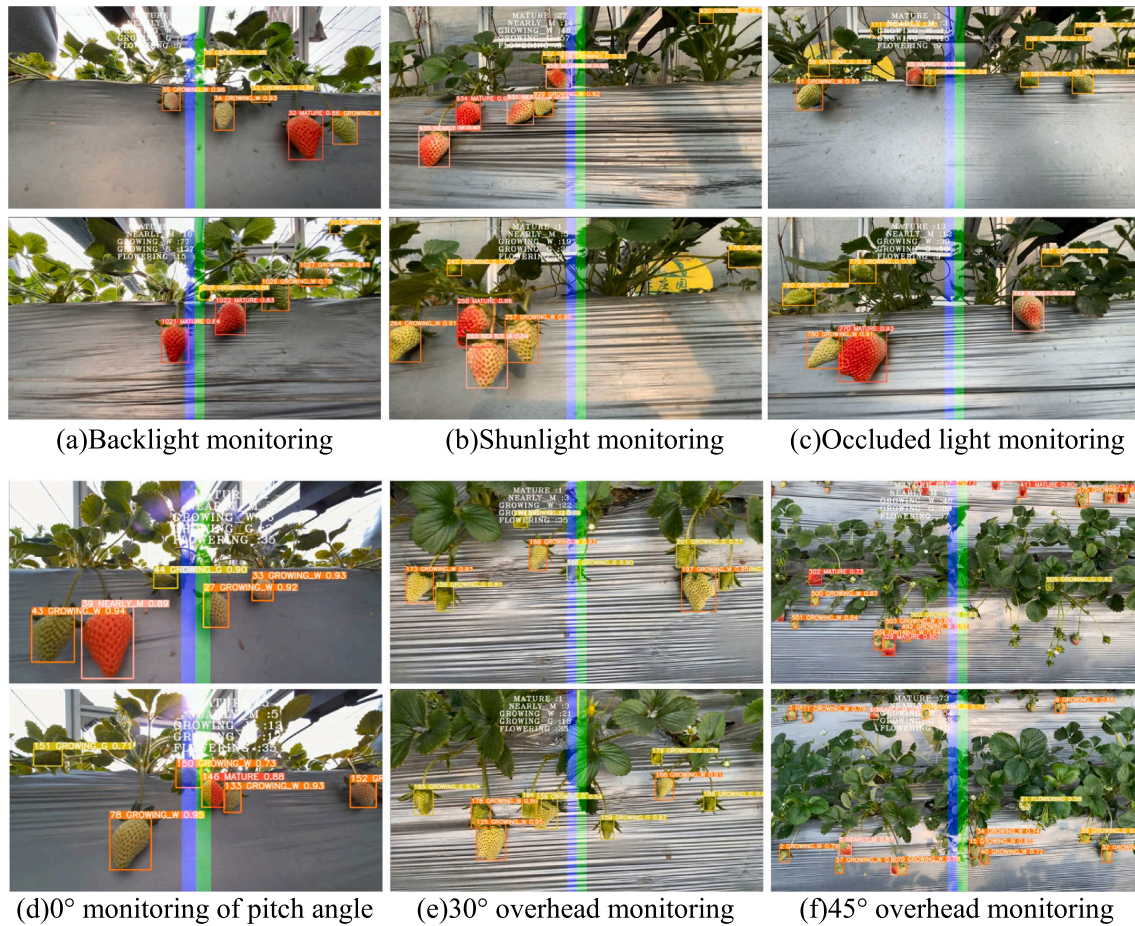


Fig. 14. Effectiveness of StraTracker monitoring under different light conditions.

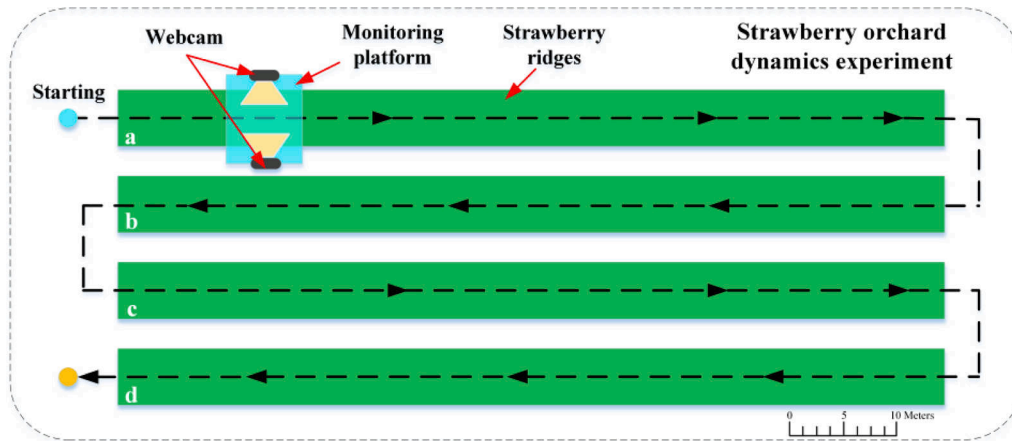


Fig. 15. Dynamic Experimental Path Planning in the Field. Figures a, b, c, and d indicate the corresponding strawberry ridges corresponding to the subsequent fruit distribution map.

time visualization of fruit distribution and growth stages was successfully achieved through stacked and occupancy maps, providing valuable insights for precise orchard management. Future work will focus on applying the model to more diverse environments and extending the StraMOT dataset to include a wider variety of strawberry varieties and orchard conditions, further improving the model's generalization capabilities.

CRediT authorship contribution statement

Qilin An: Writing – original draft, Methodology, Data curation, Conceptualization. **Yongzhi Cui:** Visualization, Validation, Investigation. **Wenyu Tong:** Supervision, Resources, Investigation. **Yangchun Liu:** Writing – review & editing, Supervision, Software. **Bo Zhao:** Supervision, Resources, Project administration. **Liguo Wei:** Writing – review & editing, Resources, Methodology.

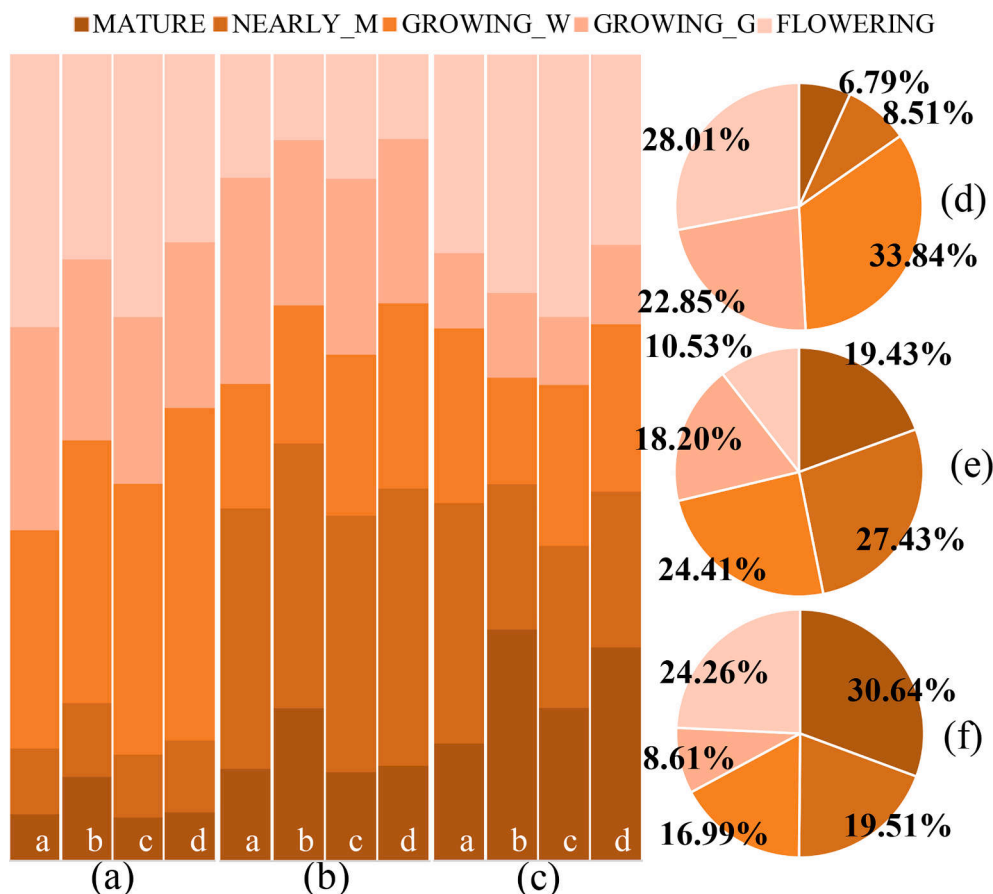


Fig. 16. Distribution of strawberries in the orchard, where figures (a) and (d) show the distribution and percentage of fruit on December 11; figures (b) and (e) show the distribution and percentage of fruit on January 2; and figures (c) and (f) show the distribution and percentage of fruit on January 10.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2024.109564>.

Data availability

Data will be made available on request.

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